

Home monitoring for the detection of atrial fibrillation in hypertensive patients. BP/AF MODE study

Monitoreo domiciliario para la detección de fibrilación auricular en pacientes hipertensos. Estudio en MODO PA/FA

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Abstract

Objective: To determine the usefulness of home blood pressure (BP) and heart rhythm monitoring strategies in the detection of subclinical atrial fibrillation (AF) to build a predictive risk score through deep learning. **Methods:** Observational, cohort, prospective, multicenter study involving 25 researchers from six Latin American countries. Home BP monitoring and single-lead electrocardiogram (ECG) recording will be performed in a population at moderate-to-high risk of developing AF. **Results:** A minimum of twenty 30-s electrocardiographic and BP recordings over 7 days using an Omron Complete Hem-7530 T ECG device will be uploaded from a mobile phone app and then sent to a database for analysis. **Conclusions:** The results of this study can provide a simple and accessible home monitoring system for detecting subclinical AF and for optimizing the predictive capacity of arrhythmia risk scores through deep learning. ClinicalTrials.gov identifier: NCT07058831.

Keywords: Hypertension. Atrial fibrillation. Home monitoring. Artificial intelligence.

Resumen

Objetivo: Determinar la utilidad de una estrategia de control domiciliario de la presión arterial y del ritmo cardiaco en la detección de fibrilación auricular subclínica para la construcción de herramientas de aprendizaje automatizado en la predic-

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*ción del riesgo de la arritmia. **Método:** Estudio observacional, de cohorte, prospectivo, multicéntrico, con la participación de 25 investigadores de seis países latinoamericanos. Se realizarán controles domiciliarios de la presión arterial y registros de electrocardiograma en una derivación, en una población de moderado o alto riesgo de desarrollar fibrilación auricular. **Resultados:** Mínimo 80% de registros electrocardiográficos válidos de 30 segundos y de presión arterial en 7 días realizados con un dispositivo Omron Complete Hem-7530 T. Los electrocardiogramas serán cargados desde una aplicación para teléfono móvil y luego enviados para su análisis a una base de datos. **Conclusiones:** Los resultados de este estudio pueden aportar un sistema sencillo y accesible de monitoreo domiciliario que permita detectar fibrilación auricular subclínica, y optimizar mediante deep learning la capacidad predictiva de las puntuaciones de riesgo de la arritmia. ClinicalTrials.gov identifier: NCT07058831.*

Palabras clave: Hipertensión arterial. Fibrilación auricular. Monitoreo domiciliario. Inteligencia artificial.

Introduction

Atrial fibrillation (AF) is a very common arrhythmia, with a prevalence, depending on the subject's gender, ranging from 3 to 5% in people aged 70-74 years and 7-10% in those aged 80-84 years. It is estimated that by 2050, one-sixth of the world's population will have AF¹.

This arrhythmia is often asymptomatic, detected incidentally during intraoperative monitoring or in patients with implanted or wireless devices, or as the first manifestation of a stroke; furthermore, in approximately 5% of cases, AF is detected within a year of the event².

It is estimated that 4% of hypertensive patients are carriers of AF, while during the evolution of the disease, another 4% will develop an episode of AF³. Subclinical AF (sAF) generally implies a lack of timely diagnosis, establishing the real possibility of the occurrence of cardioembolic stroke and peripheral arterial embolisms. Although the incidence is lower than that of clinical AF, it can generate consequences that are no less worrying in terms of years of life lost or quality of life⁴.

There are several risk scores that allow the identification of AF at-risk populations, and as stated, the technological capabilities of implanted devices allow for the detection of these asymptomatic arrhythmias.

In the last decade, there has been a significant development of wireless devices for heart rhythm monitoring. This advancement, along with the emergence of tools based on machine learning and deep learning, has sparked growing interest in the scientific community, promoting the incorporation of these emerging technologies into patient monitoring and care^{5,6}.

We propose to evaluate the usefulness of implementing a program based on obtaining 1-lead electrocardiographic recordings in patients at high risk of developing atrial fibrillation, in order to improve the detection of asymptomatic arrhythmias.

Objective

The primary objective of this study is to determine the usefulness of a home blood pressure (BP) and heart rate monitoring strategy in detecting AF in hypertensive patients in Latin America.

The secondary objective is to identify predictors of AF in a sample of Latin American hypertensive subjects at moderate to high risk of developing this arrhythmia, thereby optimizing the predictive capacity of AF risk scores.

Methods

The BP/AF MODE study is a multicenter, observational, prospective, cohort trial involving 25 researchers from six countries: Argentina, Chile, Colombia, Ecuador, Mexico, and Peru. The study was submitted to the IASC Academia (Inter-American Society of Cardiology) and approved under Project File Number 052, dated January 29, 2025. It will be conducted in full compliance with the World Medical Association Codes of Ethics for experiments involving human subjects. The Project follows the rules for diagnostic test: STARD (www.equator-network.org/reporting-guidelines/stard/)

Inclusion criteria

- Hypertensive patients of both sexes over 75 years of age.
- Hypertensive patients between 65 and 75 years of age associated with one of the following risk factors:
 - a. Obesity
 - b. Type 2 diabetes
 - c. Obstructive sleep apnea syndrome
 - d. Electrocardiogram (ECG) showing left ventricular hypertrophy
 - e. Left ventricular hypertrophy on echocardiogram
 - f. Left atrial dilation

- g. Coronary artery disease
- h. Chronic kidney disease
- i. Heart failure with ejection fraction > 50%
- j. Alcohol intake
- k. Frequent supraventricular ectopic beats.

Exclusion criteria

- History of any type of AF
- Receiving pharmacological treatments with potential impact on heart rhythm
- History of potentially malignant ventricular arrhythmias
- Uncontrolled hypothyroidism or hyperthyroidism
- Significant chronic obstructive pulmonary disease.

The diagnostic criteria for the variables that make up the inclusion criteria are detailed below.

- Definition of hypertension: patient with office BP > 140/90 mmHg and > 130/80 mmHg on 24-h ambulatory blood pressure monitoring (ABPM) or > 135/85 mmHg on daytime ABPM, or who is receiving treatment with antihypertensive drugs⁷.
- Definition of obesity: body mass index ≥ 30 kg/m² or waist circumference > 88 cm in women and > 102 cm in men.
- Definition of type 2 diabetes: glycated hemoglobin > 6.5%, fasting blood glucose > 126 mg/dL, or a 2-h glucose tolerance test > 200 mg/dL, or typical symptoms of hyperglycemia associated with a random blood glucose level > 200 mg/dL, or treatment with oral hypoglycemic agents or insulin⁸.
- Definition of Obstructive Sleep Apnea Syndrome: polysomnography with more than 5 apnea/hypopnea episodes per hour or more than 1.4 episodes/h of desaturation > 4%⁹.
- Definition of ECG with left ventricular hypertrophy: strain pattern, Sokolow-Lyon voltage criteria, Cornell, or Cornell product.
- Definition of left ventricular hypertrophy on echocardiogram: ventricular mass index > 95 kg/m² in women and 115 kg/m² in men¹⁰.
- Definition of left atrial enlargement: indexed left atrial volume on echocardiogram > 34 mL/m²⁸.
- Definition of coronary artery disease: history of acute myocardial infarction, hospitalization for acute coronary syndrome, or myocardial revascularization
- Definition of chronic kidney disease: estimated glomerular filtration rate < 60 mL/min/1.75 m².
- Definition of heart failure with ejection fraction >> 50%: clinical syndrome of heart failure with an H₂FPEF score > 4 points, and ejection fraction > 50%

by volumetric echocardiographic method, magnetic resonance imaging, or radiocardiogram¹¹.

- Definition of cerebrovascular disease: history of paresis, paralysis, or amaurosis fugax lasting more than 30 min, documented in clinical history or imaging studies with cerebral, cerebellar, or brainstem infarction or hemorrhage.
- Definition of peripheral vascular disease: history of amputation due to critical ischemia, intermittent claudication associated with stenosis > 50% in at least one territory, or an ankle-brachial index < 0.9 in the affected lower limb.
- Definition of alcoholic beverage intake: intake of more than four 75 mL glasses of wine or equivalent per day.
- Definition of frequent supraventricular ectopic beats: more than 10 supraventricular ectopic beats per hour on a 24-h Holter ECG.

Cardiovascular risk stratification

The HEARTS cardiovascular risk estimator will be used, which includes gender, age, history of ischemic heart disease, cerebrovascular disease, or peripheral vascular disease, chronic kidney disease, diabetes mellitus, smoking, total cholesterol, and systolic BP. Patients with a history of ischemic heart disease, cerebrovascular disease, or peripheral vascular disease do not require stratification, as they are considered very high risk, as are subjects with chronic kidney disease or diabetes mellitus, who are considered high risk¹².

High to moderate risk of developing AF

The risk of developing new AF was determined through the HARMS2-AF and modified Taiwan AF scores. The HARMS2-AF score includes the variables hypertension, age, raised body mass index, gender, sleep apnea, smoking, and alcohol consumption. Each variable receives a value, and the sum of them could be from 0 to 14 points. The modified Taiwan AF score includes the variables age, gender, hypertension, heart failure, coronary artery disease, and end-stage renal disease. The final sum could be from –2 to 14 points. HARMS2-AF score > 5 or a modified Taiwan AF score > 4 is considered moderate-to-high risk of developing AF^{13,14}.

Frailty assessment

Frailty will be assessed using the FRAIL index. It includes five domains: weight loss, exhaustion,

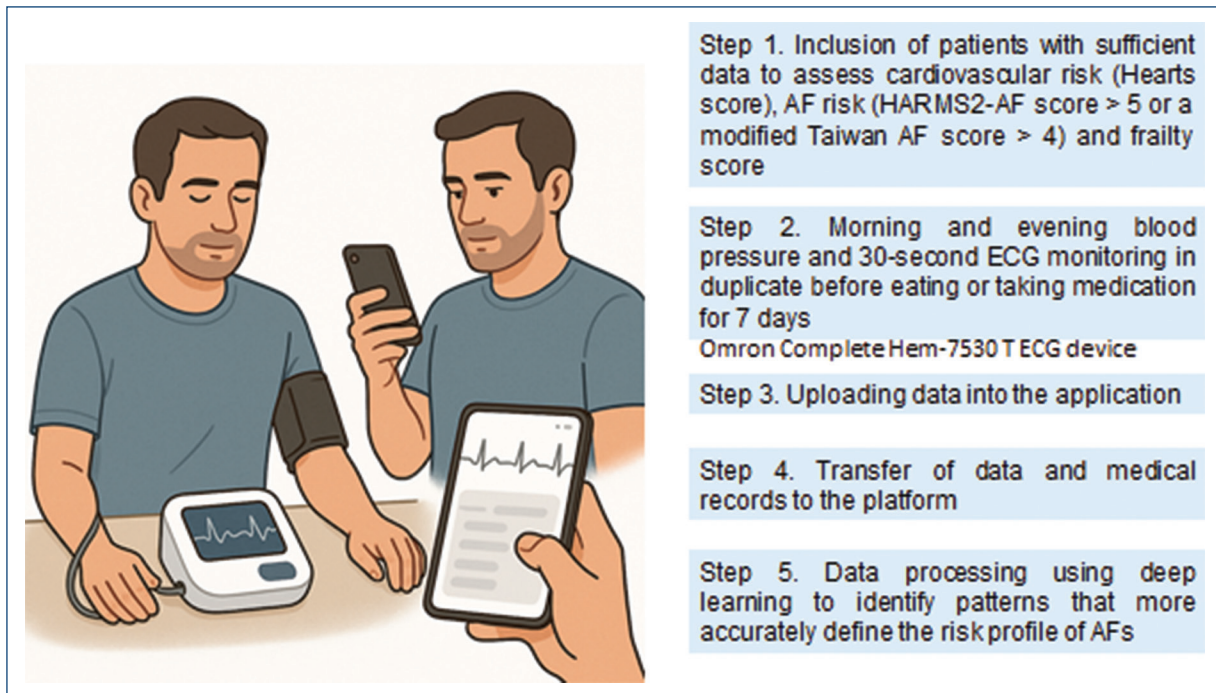


Figure 1. PA/AF MODE study procedures.

weakness, slow gait, and reduced physical activity. The score ranges from 0 to 5, with 0 being non-frail and 5 being extremely frail. Subjects with a score of 0 were classified as “non-frail,” 1-2 as “pre-frail,” and ≥ 3 as “frail”¹⁵.

Procedures

The 30-s home ECG tracings will be obtained using the Omron Complete Hem-7530 T ECG device (Omron Healthcare Co., Ltd., Japan). The connection and smartphone app download will be available from Omron Connect free of charge on the Google Play Store: https://play.google.com/store/apps/details?id=jp.co.omron.healthcare.omron_connect&hl=es_419&pli=1 and on the iOS App Store: <https://apps.apple.com/mx/app/omron-connect/id1003177043>. Two duplicate ECG recordings will be required daily: the first in the morning before breakfast and taking the prescribed medication, and another in the evening before dinner and taking the afternoon/evening medication. Additional ECG recordings will be required if the patient experiences any symptoms. The study investigator will work with the patient to download the app, perform the first ECG tracing, and transmit it to the database. The patient will be given instructions on how to correctly perform the ECG and transmit it to the platform, along with a patient

diary. Simultaneously with obtaining the ECG tracings, the patient will take two BP measurements, in a seated position, separated by 1 min, following the recommendations of the Latin American Society of Hypertension guidelines⁵. After 1 week, the patient will return the device to the research center. A recording will be considered valid when at least 80% of the ECG tracings and BP measurements are valid (Fig. 1).

Statistical analysis

Considering those patients with a modified Taiwan AF score > 5 points, whose annual incidence of AF ranges between 1.54 and 6.98%, or a HARMS2-AF score > 7 points, whose annual incidence of AF ranges between 1.35 and 6.62%, it will be required to include 499 patients to achieve a 5% margin of error, with a 99% confidence level. If the sample calculation does not achieve the expected results, recruitment will continue until 25 cases of AF are reached.

Continuous variables will be reported as means with their standard deviations for normally distributed variables, or median and interquartile range for non-normally distributed variables. Discrete variables will be reported as absolute values and percentages. Unpaired ANOVA will be used to analyze normally distributed

variables, whereas the Kruskal-Wallis test will be used for non-normally distributed variables. Differences in proportions will be assessed using the Chi-square test. A two-tailed $p < 0.05$ will be considered statistically significant.

Discussion

Without any doubt, healthcare is facing a paradigm shift in patient care as a result of the shift toward the digital era, driven by technological advances already available to the global community. Cardiovascular diseases are not immune to these changes, forcing healthcare teams to incorporate new wireless devices through programs that include patient training and, therefore, foster their empowerment.

A first question regarding AF detection is which patient population can benefit most, and it is clear that this will be those at greatest risk.

Although hypertension is perhaps the most important risk factor for developing AF, it is not the only one. The HARMS₂-AF risk score for developing a new arrhythmia includes age over 60 years, obesity, male gender, the presence of obstructive sleep apnea syndrome, smoking, and consuming at least seven standard servings of alcoholic beverages per week¹⁰. Individuals with a HARMS₂-AF score between 5 and 9 points increase their risk of developing a new AF by 12.79 times (95% confidence interval [CI]: 8.93-18.33), and with a score of 10-14 points, the risk increases by 38.70 times (95% CI: 26.96-55.54). The area under the curve for the score was 0.782, higher than the Framingham-AF model of 0.568.

The modified Taiwan AF score considered age, male sex, the presence of hypertension, heart failure, coronary artery disease, and chronic kidney disease, and excluded alcoholism from the original score in the risk stratification of AF. The group of patients with a score of 1 or 2 points had an annual incidence of arrhythmia of 0.2%, in those subjects with a score between 4 and 9 points, the incidence was 1.33% per year, whereas in those with a score of 9-14 points the incidence increased to 3.6% per year¹¹. The area under the curve to detect new AF was 0.861 in the 1st year of follow-up, progressively decreasing to 0.751 at 16 years of follow-up.

In this scenario, it is assumed that the use of algorithm platforms may improve risk characterization, especially when applied to a specific population, which may be different from those used in the literature to define prevalence and incidence rates.

A second aspect to consider is whether low-complexity equipment can provide important information for identifying sAF.

The RITMO study was a program to optimize the detection of paroxysmal AF. Subjects over 65 years of age with hypertension or heart failure and no symptoms compatible with arrhythmias were included. AF risk was stratified using the Stroke Risk Analysis (SRA) using a 1-h ECG strip. Patients classified as high risk were prescribed a 7-day home monitoring protocol with the Kardia 6 device (OMRON, AliveCor®). 2.8% of patients with hypertension were detected with AF using the SRA, and 10.2% of new cases were detected during home ECG monitoring¹⁶.

In a project that included diabetic patients in primary prevention, an automatic device was used to measure BP and detect AF, Microlife WatchBP Office AFIB. The prevalence of AF was 1.9% (95% CI 1.3-2.6), increasing from 0.5% in patients younger than 65 years to 2.2% in patients between 65 and 74 years, and to 4.3% in those older than 75 years. The authors concluded that AF screening with this type of device can be considered a standard procedure in this type of clinical situation¹⁷.

The BP/AF MODE study proposes an automated BP monitor with the ability to record ECGs in one lead for 30 s. The study population is at moderate-to-high risk for AF.

Conclusions

The purpose of this publication is to document an out-of-office algorithm to detect sAF. The results of this study can provide a simple and accessible home monitoring system for detecting sAF and optimizing the predictive capacity of arrhythmia risk scores through deep learning.

The usefulness of a BP and heart rate monitoring system, with patient or caregiver participation and low-cost equipment, to detect AF cases will be established. Deep learning tools could improve risk profile definition beyond the capabilities of the scores used so far.

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Conflicts of interest

The authors declare no conflicts of interest.

Ethical considerations

Protection of humans and animals. The authors declare that the procedures followed complied with the ethical standards of the responsible human experimentation committee and adhered to the World Medical Association and the Declaration of Helsinki. The procedures were approved by the institutional Ethics Committee.

Confidentiality, informed consent, and ethical approval. The authors have followed their institution's confidentiality protocols, obtained informed consent from patients, and received approval from the Ethics Committee. The SAGER guidelines were followed according to the nature of the study.

Declaration on the use of artificial intelligence. The authors declare that no generative artificial intelligence was used in the writing of this manuscript.

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